Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

It will use for loops to iterate over elements in chemical equations and multiple if conditionals for logic in determining how to handle each element and figure out what it is. I want to get better at composing multiple functions together throughout this course. I feel that I'm good at writing individual functions that work but putting them together in a way that makes sense is difficult.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think determining what letters represent elements will be difficult because some consist of 2 letters while others are 1. Also polyatomic ion/parenthesis will be difficult to process since they have to be multiplied in the correct order. I haven't figured out if this will be possible with my knowledge yet, or how to do it.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The processing part will take the most effort. I want to implement the user input part first, since it will be simple and I will have input variables to reference when making the processing part. By doing the input first, the processing will be less abstract to code and thus easier.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Overwatch competitive grind. I'll minimize it by remembering this project affects my grade and is more important, and also time sensitive. I will need the drafter and python official documentation.

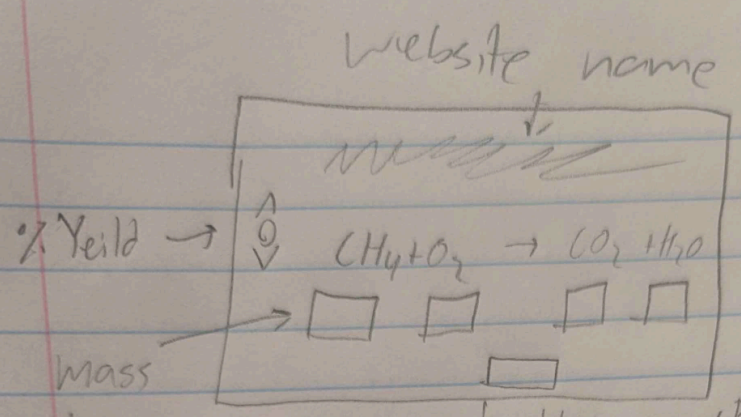
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.



State:

equation: str

balanced: list[str]

mass1: float

mass2: float

mass3: float

mass4: float

yield: float

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	to balance chemical reactions. Ultimately to save time by doing math automatically for those who already understand the fundamentals behind it.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	reactants: str products: str balanced: list[str] the reactants and products represents raw user input as a string. the balanced field will be filled once the computer balances the equation.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Polyatomic element_1: str count_1: int element_2: str count_2: int It will help keep track of polyatomic ion logic. I can create a function that extracts this data out of a string and turns it into a dataclass instance which can be used easier in calculations.

Routes (minimum 7) The names of your route functions	updateYield calculate updateMass index showResults showTable
Pages (minimum 5) A brief description of the pages in your site	index - the base page updateMass - the index if the user updates a mass number with new calculations that reflect this updateYield - the index if the user updates %yield and new calculations that reflect this showResults - the final page once the computer calculated results table - a popup containing the periodic table as reference so users can enter the elements correctly
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	TextBox - chemical reactants TextBox - chemical products Button - when the user is ready to submit data and have the computer calculate TextBox - percent yield TextBox(0-4) - a textbox created for every reactant/product the user enters, which optionally lets them enter a mass
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	I will have to use multiple “if not”s since there are some fields that are optional and have to be handled correctly when left empty. For example, masses and percent yield. I will use an if statement to determine if a letter in a chemical equation is lowercase or uppercase. By doing this I can determine if an element is 1 or 2 or 3 characters long.

Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	A for loop will loop over the balanced list[str] in my State. I will have to continually loop over it to balance the coefficients and produce the correct outcome.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Didn’t use an LLM

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.